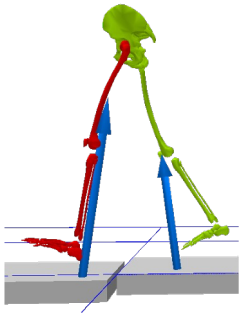
SM2501/FSG3084

Lecture 8

Kinetics of normal gait



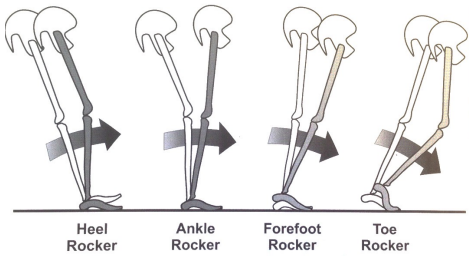


1

Normal Gait, *review*

Prerequisites for Normal Gait (Gage 1991)

- Stability in stance
- Foot clearance in swing
- Pre-positioning of foot for initial contact
- Adequate step length
- Energy conservation

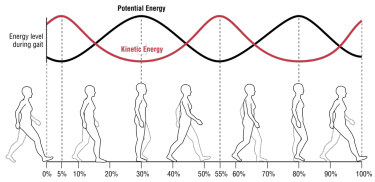


Foot rockers, Jacquelin Perry, 2010

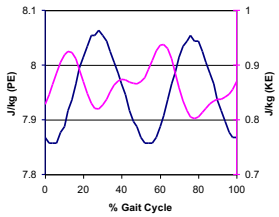
2

"65% recovery of mechanical energy through pendular motion"

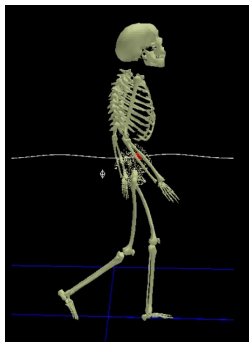
Cavagna et al. 1976, Dipaola et al. 2016, Bennett et al, 2005



Neumann 2002



Experimental observation



Energy cost of walking

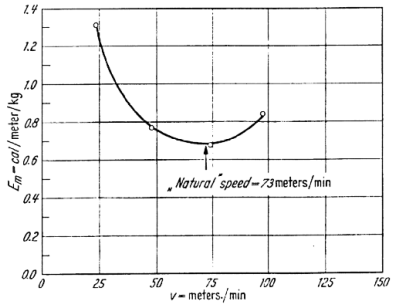
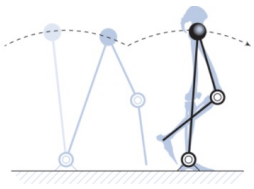


Fig. 3. Normal male subject. Relation between energy expenditure in cal/meter/kg and speed. Arrow indicates the "natural" walking speed of subject

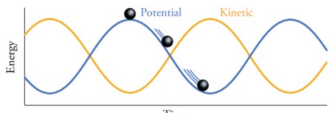
Ralston 1958

3

Walking without muscles? -> Passive walkers



Delp & Uchida 2020



http://www.youtube.com/watch?v=kfVJyJrMf38

http://www.youtube.com/watch?v=HSKyHmjyrkA

Many more....

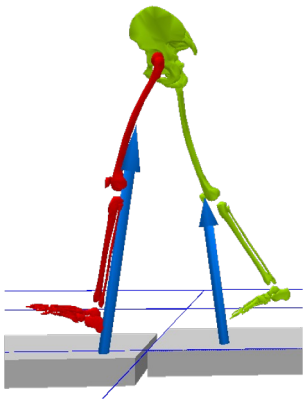


5

Kinetics of normal gait

Why study kinetics?

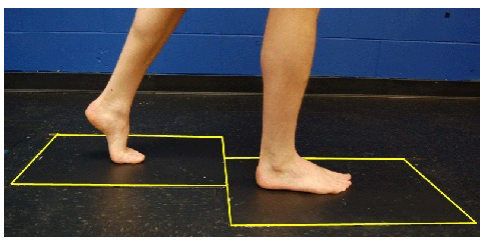
- Kinetics make it possible to study mechanisms for
  - Controlling motion
  - Producing motion
- Provides a different perspective than kinematics
- Can NOT be observed by the naked eye

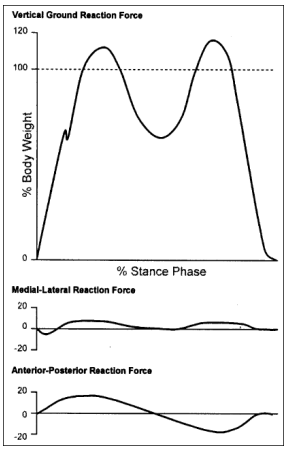


6

Ground reaction force measurement

Force plates: F_x, F_y, F_z, M_x, M_y, M_z, Center of Pressure



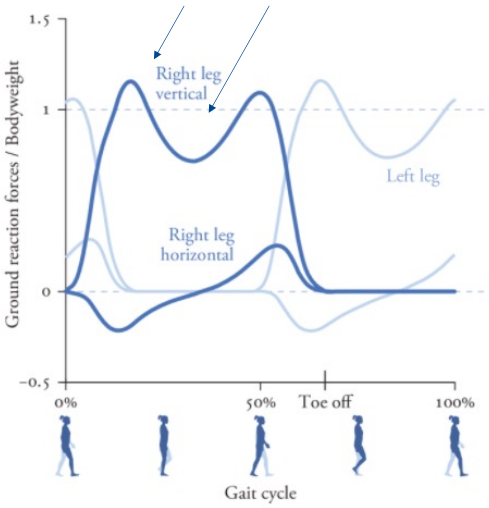


Perry 2004

7

Kinetics

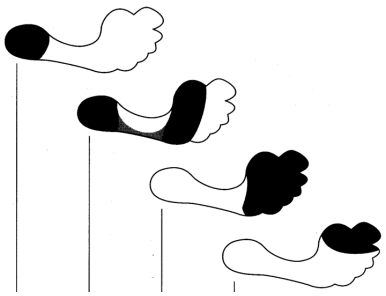
Vertical direction of COM acceleration?

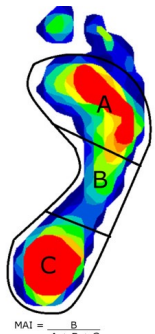


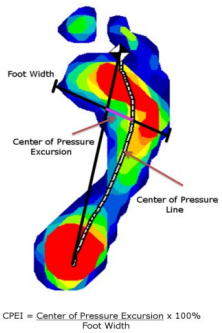
Polygon

8

Loading under foot during gait



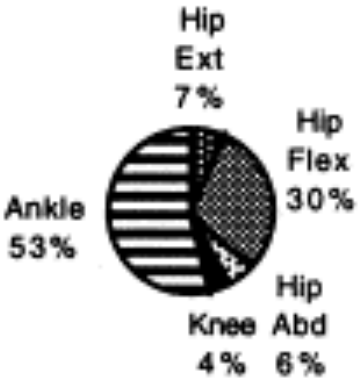




Center of Pressure progression

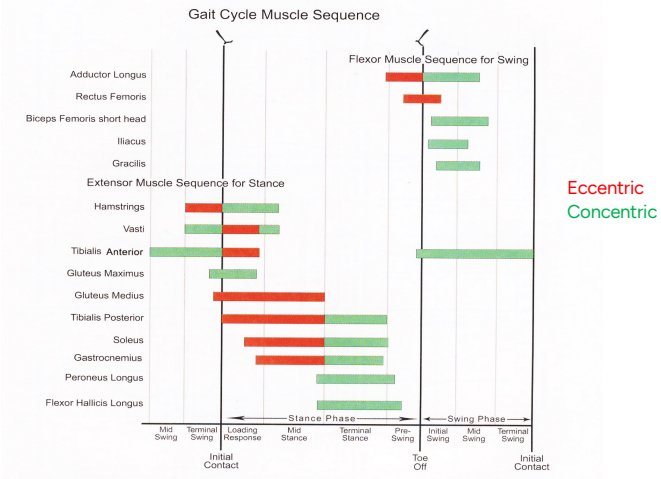
9

Contributions to positive work



10

Muscle activity (EMG timing)



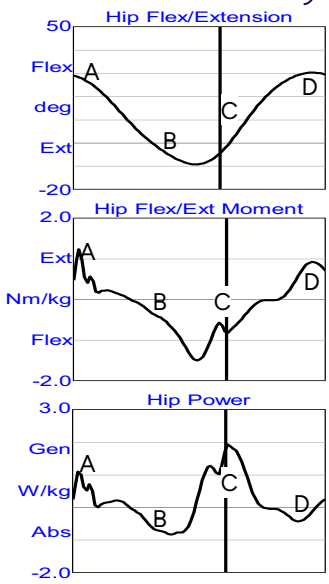
Eccentric
Concentric

Gage 2009

11

Hip

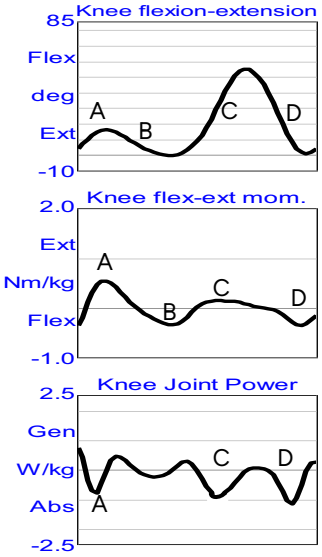
- A:
- B:
- C:
- D:



12

Knee

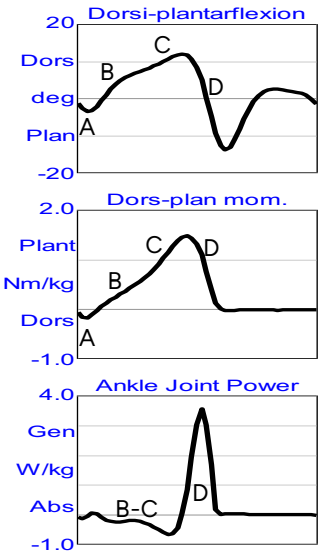
- A:
- B:
- C:
- D:



13

Ankle

- A:
- B:
- C:
- D:



14